

Path for Forensic Examiner Aces

Tuan Phantastic

CornCon Cybersecurity Conference

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#Digital Forensics #DFIR

Introducing: Tuan Phantastic

Tuan works for a big well-known Financial institute...



As an Assistant Vice President (AVP) of eDiscovery and Forensics Examiner Lead and has many years of hands-on technical experience, including EF/DF/DFIR, and Insider Threat Strategy.

He is a Principal Security Engineer for

ThreatReel

<https://threatreel.com>

Follow / contact Tuan:

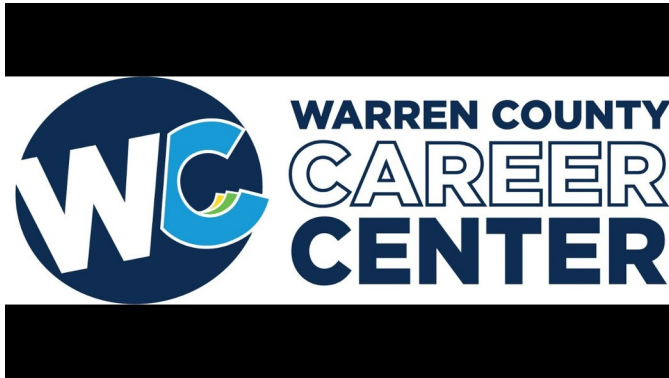
- <https://www.linkedin.com/in/tuanqphan>
- <https://github.com/phanrensicss/slides>



Where I Volunteer...



Technical
Mentor



Advisory Board Member

Disclaimer!

Yes, the presenter have day jobs. However...

Opinions expressed are based solely on his own independent security research and do not express or reflect the views of his employers.



Agenda

What is a Digital Forensics Examiner (DFE)

- Baseline knowledge required for DFE
- Public vs Private
- Basic technical skills and the experience
- Q&A



“Forensic”



White Lab Coat



Clean Room\Forensic Lab

Why This Talk?

- This talk is inspired by job candidates I participated in interviewing.
- Help fill the knowledge, clear the confusion and skill gaps.
- Digital Forensics – Is this the career path for you?

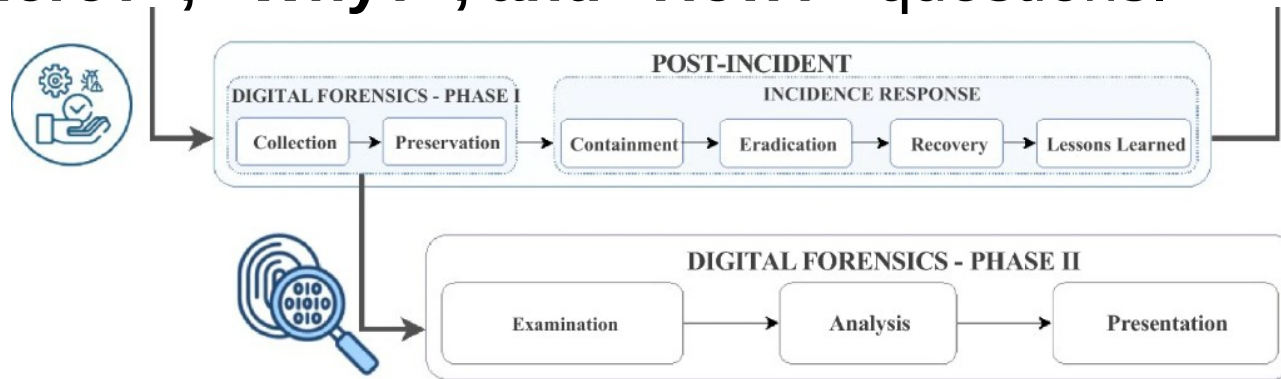
Digital Forensics

- **The process of identifying, collecting, preserving, extracting, analyzing, and presenting evidence.**
 - Manner that is legally acceptable, presented in a Court of Law or other.
- **Data is collected without alteration,**
 - Every step can be traced for legal and compliance purposes.
- **Reconstruction of events provide evidence of crimes or case of user misconduct.**
- **Data and Metadata**
 - Author\Owner
 - Creation date
 - Modify date
 - Last Access date
 - Printed date



Roles of DF\DFIR

- **Digital Forensics Examiner or Digital Forensics Incident Response (DFIR)**, both investigate security incidents, events, and alerts to answer the classic “**Who?**”, “**What?**”, “**When?**”, “**Where?**”, “**Why?**”, and “**How?**” questions.



- **DF** essentially the collection, **preservation**, and analysis of data.

Aspiring DFE

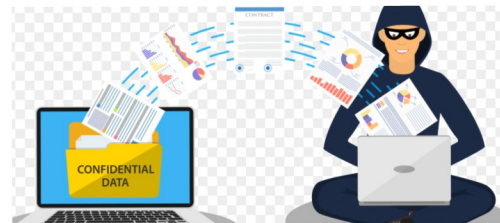
- Retain a working level knowledge of the 5 steps of digital forensics.
 - 1) Identification
 - 2) Data collection\preservation
 - 3) Analysis & Examination
 - 4) Reporting\Presentation
 - 5) ***Documentation**



When Digital Forensic Starts

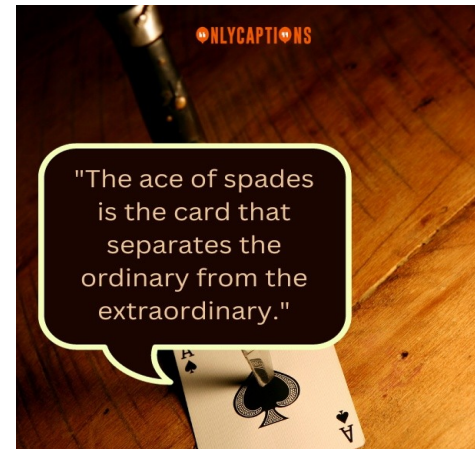
Tied to some kind of criminal or Cyber crime investigation.

- Any events that happen in your company's perimeter.
 - Inappropriate conduct\Fraud
 - Data being tampered\thief
 - Misuse of company resources
 - Someone getting access to something that they shouldn't have had or in a malicious way.
 - Mouse jigglers



What Does it Takes to be a DFE

- 1) **Analytical Thinking:** Ability to piece together clues and draw conclusion from complex data.
- 2) **Attention to Detail:** Precision is critical when handle and analyzing evidence.
- 3) **Communication Skills:** Teamwork and willingness to communicate and work well with others.
- 4) May work long hours.
- 5) Ability to be good under pressure.
- 6) *Testify in court.



Public vs Private

Public Sector

- Work with regional government bureaus
- Deal most with devices for evidence of criminal activities such as hacking, fraud, homicide, Drugs enforcement, Child abuse, and others.
- Goal is to provide evidence of crime.
- Lab Accreditation for ISO 9001

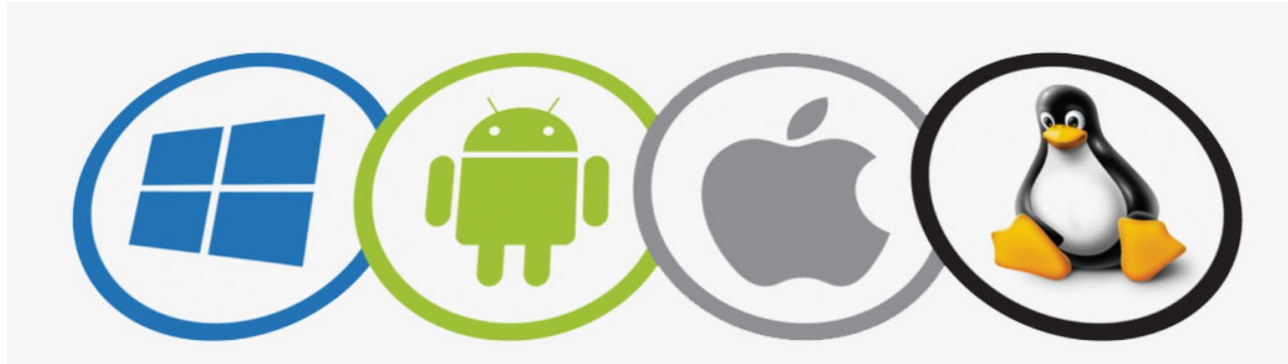


Private Organization

- Deal the same types of device analyses as public, but less intense in mobile volume.
- Email archives, SIEM, EDR, Compliance Portal, DLP, and other enterprise tools.
- Legal Right Hold (LRH)\eDiscovery -
 - FRCP Rule 37
- Goal is to solve the case of the user and prevent potential future threats.
- Not all lab are Accreditation for ISO 9001

OS Basic Skills

- **General knowledge of operating systems;**
 - Windows,
 - Linux & Mac
 - IOS
 - Android
 - Others



Basic Artifacts Skills

- **Windows:**

- Recycle Bin/Deleted data, Registry, Browser artifacts, File Activity, winevt, SRUM, and others.

- **Linux & Mac:**

- Trash Bin, File System Events, Browser artifacts, File Activity, *.plist, Recently Used Items, KnowledgeC: Application Focus, and others

- **Mobile devices (iOS\Android):**

- Geo-location, communication, application, media, web browser, and others

File-system Basic Skills

Knowing where things are supposed to be located on an endpoint\devices

Windows

- C:\Recycle.bin
- \Windows\System32\Config*registry
- \Windows\System32\sru\SRUDB.dat
- \Users\AppData\Local\ConnectedDevicesPlatform\<id>\ActivitiesCache.db

Linux\Mac

- /home/<USERNAME>
- /etc, /var/log, and /etc/passwd
- *.plist

IOS

- /mobile/Library/Safari/
- /mobile/Library/SMS/sms.db
- /mobile/Library/CallHistoryDB/

Android

- /system/packages.list
- /system_ce/0/recent_images
- /data/system/usagestats/0

<https://www.sans.org/posters/windows-forensic-analysis/>

Cloud Storage

OneDrive

Description

OneDrive is installed by default on Windows 8 systems, although it must be manually installed on other operating systems. It is the default cloud storage service for Windows 8.

Location

Windows 8
C:\Users\user\OneDrive

OneDrive folder location

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Google Drive for Desktop

Description

Google Drive for Desktop is the desktop client for Google Drive. It allows users to sync files and folders from their Google Drive account to their local hard drive.

Location

Google Drive for Desktop is installed in the Program Files folder. The default installation path is C:\Program Files\Google\Drive\GoogleDrive.exe.

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<https://www.sans.org/posters/dfir-advanced-smartphone-forensics/>

[illegible][illegible]

*Important Skill: Logs

- The bedrock and starting point of most security investigations begins with any available logs.
- Private vs Public



Native Logs

Understanding where logs exist on various endpoints is vitally important

- **Windows Logs**

- %System32%\winevt\Logs

- **Linux Logs**

- /var/log/syslog

- **Server Logs**

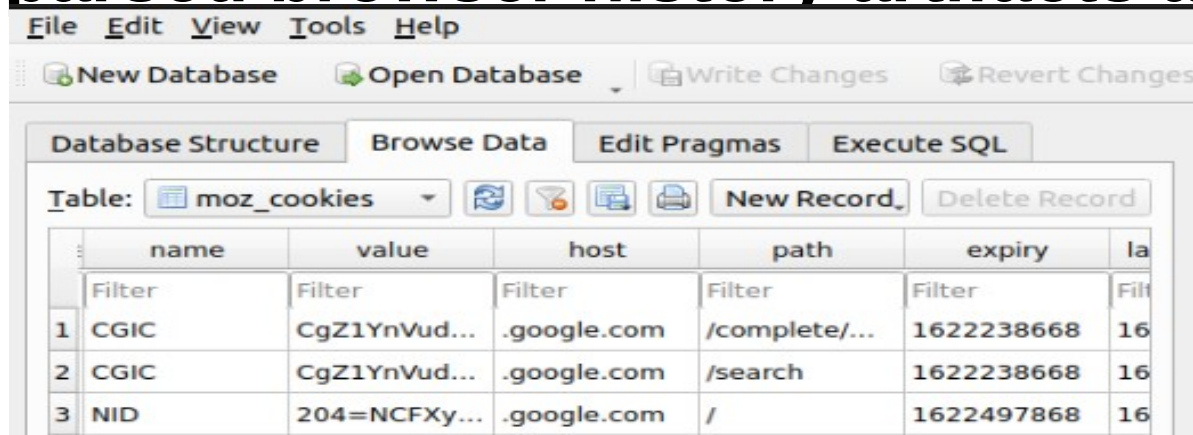
- %SystemDrive%\inetpub\logs\LogFiles

- **Application Logs**

- Depends on applications, some may stored in centralized management

SQL Basics Skills

- Understand “Structured Query Language” (SQL) to parse browser history artifacts and other *.DB



The screenshot shows a database browser window with a menu bar (File, Edit, View, Tools, Help) and a toolbar with buttons for 'New Database', 'Open Database', 'Write Changes', and 'Revert Changes'. Below the toolbar are tabs for 'Database Structure', 'Browse Data', 'Edit Pragmas', and 'Execute SQL'. The 'Browse Data' tab is active, showing a table named 'moz_cookies'. The table has columns: name, value, host, path, expiry, and last access time (labeled 'la' in the screenshot). The table contains three rows of data, all with a 'Filter' button in the first column.

	name	value	host	path	expiry	la
	Filter	Filter	Filter	Filter	Filter	Filter
1	CGIC	CgZ1YnVud...	.google.com	/complete/...	1622238668	16
2	CGIC	CgZ1YnVud...	.google.com	/search	1622238668	16
3	NID	204=NCFXy...	.google.com	/	1622497868	16

- Free online training
 - <https://www.sqlcourse.com/beginner-course/>
 - <https://www.sqlcourse.com/advanced-course/>

File Type Identification Skills

- **Understand file signatures, file headers, or whatever you prefer to call them...**
 - Because file extensions can be altered or removed
- **https://www.garykessler.net/library/file_sigs.html**

File Type Case Study

The screenshot displays a file explorer window with a directory named 'miscellaneous'. The directory contains the following files:

File Name	Extension	Size	Type
201702100_notes.txt	.txt	5,961	Document
211.txt	.txt	1,405,904	Document
8s.txt	.txt	4,856,910	Document
a.txt	.txt	15,961,124	Document
important.txt	.txt	3,833	Document
samplecode_sas_class.txt	.txt	523,242	Document
sas_ole_db_instructions.txt	.txt	6,495	Document
sqa.txt	.txt	1,652,495	Document
timkehoe.txt	.txt	190	Document

The file '211.txt' is selected, and its raw data is displayed in the hex editor view below. The first few bytes of the file are circled in red, showing the following hex values:

```
00000000 PK.....x0J)$.±.v..pu...2015+Tony+Awards+Arrivals+I123zdpdloj1.jpg
```

Identify File Execution

- Understand how to know if a file executed on a Windows endpoint
 - Some Examples for the registry key:
 - **UserAssist**
 - **RecentApps**
 - **ShimCache**
 - **CurrentVersion**
 - **Prefetch** - %windir%\Prefetch

File Execution Case Study

- UserAssist

- How many times was the File Explorer launched? (26)

Registry hives (1) Available bookmarks (31/0) Values UserAssist

Enter text to search... Find

Key name

Enter text to search... Find

Drag a column header here to group by that column

Program Name	Run Counter	Focus Count	Focus Time	Last Executed
UEME_CTLCUACount:ctor	=	=	0 0d, 0h, 00m, 00s	
{Common Programs}\Accessories\Snipping Tool.lnk	9	0	0d, 0h, 00m, 00s	2021-11-25 03:14:34
UEME_CTLSESSION	54	0	0d, 0h, 00m, 00s	
{Common Programs}\Accessories\Paint.lnk	7	0	0d, 0h, 00m, 00s	2021-11-25 03:14:34
{Programs}\Accessories\Notepad.lnk	6	0	0d, 0h, 00m, 00s	2021-11-25 03:14:34
{User Pinned}\TaskBar\File Explorer.lnk	26	0	0d, 0h, 00m, 00s	2021-12-01 13:02:43
{Programs}\Windows PowerShell\Windows PowerShell.lnk	1	0	0d, 0h, 00m, 00s	2021-11-25 03:37:24
{User Pinned}\TaskBar\Firefox.lnk	2	0	0d, 0h, 00m, 00s	2021-12-01 12:32:34
{Common Programs}\Accessories\Remote Desktop Connection.lnk	1	0	0d, 0h, 00m, 00s	2021-11-25 03:59:55
{User Pinned}\TaskBar\Opera Browser.lnk	1	0	0d, 0h, 00m, 00s	2021-11-25 04:10:02
{Common Programs}\Accessories\Notepad.lnk	1	0	0d, 0h, 00m, 00s	2021-11-30 10:55:21

Identify File Activity Skill

- Understand how to know if a file was accessed.
 - This activity is tracked under the following registries below:
 - Jump Lists
 - RecentDocs
 - LastVisitedMRU – last path file opened\executed
 - OpenedSavedPidMRU
 - LNK files
 - ShellBags
 - etc

File Activity Case Study

- RecentDocs

Registry hives (1)				Available bookmarks (21/0)						
Enter text to search...				Find						
Key name	# values	# subkeys	Last write timestamp							
HKEY_CURRENT_USER	=	=	=							
FeatureUsage	1	3	2020-09-29 15:35:01							
FileExts	0	185	2020-09-29 17:27:01							
HideDesktopI...	0	1	2020-09-29 15:07:50							
LogonStats	2	0	2020-09-29 15:04:53							
LowRegistry	0	0	2020-09-29 15:04:55							
MenuOrder	0	1	2020-09-29 15:04:55							
Modules	0	3	2020-09-29 15:24:27							
MountPoints2	0	3	2020-09-29 15:24:31							
Package Inst...	1	0	2020-09-30 14:32:10							
RecentDocs	5	3	2020-09-30 14:40:31							
Ribbon	2	0	2020-09-29 15:24:21							
RunMRU	0	0	2020-09-29 15:24:29							
SearchPlatform	0	1	2020-09-29 15:04:52							
Shell Folders	31	0	2020-09-29 15:04:56							
Shutdown	1	0	2020-09-30 14:43:30							

Values						
Recent documents						
Drag a column header here to group by that column						
Extension	Value Name	Target Name	Lnk Name	Mru Position	Opened On	Extension Last Opened
HKEY_CURRENT_USER	HKEY_CURRENT_USER	HKEY_CURRENT_USER	HKEY_CURRENT_USER	=	=	=
RecentDocs	3	temp	temp.lnk	0	2020-09-30 14:40:31	2020-09-30 14:40:31
RecentDocs	2	users.txt	users.lnk	1		2020-09-30 14:40:31
RecentDocs	0	The Internet	The Internet.lnk	2		
RecentDocs	1	kgjcheck/	ms-gamingoverlay--kgjcheck-.lnk	3		
Folder	1	temp	temp.lnk	0	2020-09-30 14:40:31	
Folder	0	The Internet	The Internet.lnk	1		
.txt	0	users.txt	users.lnk	0	2020-09-30 14:40:31	

Identify USB Activity Skill

- **Understand how to know if a USB was connected.**
 - **The majority of USB-related artifacts are located within the Windows Registry**
 - SYSTEM\CurrentControlSet\Enum\USBSTOR
 - SYSTEM\MountedDevices
 - HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist
 - **Windows Event Logs:**
 - System event logs may contain entries related to USB device insertions and removals.
 - Event Viewer path: Applications and Services Logs\Microsoft\Windows\DriverFrameworks-UserMode\Operational

USB Activity Example

- **USBSTOR:**

- SYSTEM\CurrentControlSet\Enum\USBSTOR

- **Things to Record:**

- Device S/N:
 - Vendor:
 - Product:
 - Service:
 - Container Id:

Value Name	Value Type	Data	Value Slack
%c	%c	%c	%c
DeviceDesc	RegSz	@disk.inf, %disk_devdesc%;Disk drive	00-00-00-00
Capabilities	RegDword	16	
Address	RegDword	2	
HardwareID	RegMultiSz	USBSTOR\Disk_USB____SanDisk_3.2Gen11.00 ...	00-00-00-00
CompatibleIDs	RegMultiSz	USBSTOR\Disk USBSTOR\RAW GenDisk	
ClassGUID	RegSz	{4d36e967-e325-11ce-bfc1-08002be10318}	00-00-00-00-00-00
Service	RegSz	disk	00-00
Driver	RegSz	{4d36e967-e325-11ce-bfc1-08002be10318}\0001	00-00-00-00
Mfg	RegSz	@disk.inf, %genmanufacturer%;(Standard disk d...	00-00-00-00-00-00
FriendlyName	RegSz	USB SanDisk 3.2Gen1 USB Device	00-00
ConfigFlags	RegDword	0	
ContainerID	RegSz	{3deb34e4-837d-5c67-b987-c26b48b8df68}	69-00-63-00-65-00

Understand Order of Volatile

- **Collect and Protect information relating to an incident**
 - Many different data source and protection mechanisms
- **RFC 3227 – Guidelines for Evidence Collection and Archiving**
 - A good set of best practices
- **How Long Does data stick around?**
 - Some media is much more volatile than others
 - Gather data in order from most to less
 - registers, cache
 - routing table, arp cache, process table, kernel statistics, memory
 - temporary file systems
 - disk
 - remote logging and monitoring data that is relevant to the system in question
 - physical configuration, network topology
 - archival media

Forensics Tools for skill building

- Many tools can be used to perform data analysis on different Operating Systems.
- Forensic toolkit for Win/Linux
 - Kali, Helix, DEFT, Autopsy Sleuth-kit, SIFT
 - FTK imager, Nirsoft
 - Velociraptor, Kape
- Mobile Forensics
 - SAFT, Autopsy, Belkasoft
 - iLEAP/aLEAP



Ways to build Forensic skills

- **Follow Cyber Security Linkedin group free training and webinars.**
 - CYBER SECURITY FORUM INITIATIVE – CSFI
 - Forensic Focus
 - Belkasoft
 - Sleuth Kit Labs
- **Follow and watch YouTube Digital Forensics Channel**
 - TEDx Talks
 - SANS
 - Google Career Certificates
 - Others



Just Some of the Challenges

- Rapidly changing technology and Data volumes
- System & Application update
- Mental health
- Ongoing education
- Unsupported devices
- Encryption\Security Features



Certifications

- **Some well-known forensic certifications include the following:**
 - **CISSP** – Certified Information Systems Security Professional
 - **CCE** – Certified Computer Examiner
 - **CFCE** – Certified Forensic Computer Examiner
 - **GIAC** – Certified Forensic Analyst (GCFA)
 - **CompTIA A+, Network+, Security+**

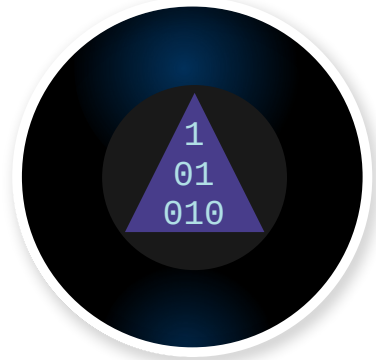
Salary & Benefits

- **The starting salary for a computer forensics professional depends on various factors.**
 - Whether you're employed in the public or private sector.
 - Degree (s) vs Certification (s)
 - # of years of experience
 - Location

Questions



Who?
What?
When?
Where?
Why?
How?



Thank you for Attending!



Phantastic!



<https://github.com/phanrensic/slides>